ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** the covered half remains brown with iodine	Light needed.
2	**(c)** returns along the same path	For 90°, reversed.
3	**(c)** more extreme	Larger variation.
4	**(d)** light energy to chemical energy	Stored in glucose.
5	**(d)** umbra	Complete shadow.
6	**(b)** more frequent leap years	Larger quarter.
7	**(b)** nitrogen-fixing bacteria	Rhizobium.
8	**(b)** virtual image position follows the law of reflection	Geometric consequence.
9	**(a)** the Earth always has a shadow but the Moon enters it only during eclipses	Shadow is always there.
10	**(c)** chlorophyll is needed	Pigment.
11	**(d)** increases by 15°	Twice the turn of the mirror (10 x 2).
12	**(b)** 12 hours	Half.
13	**(a)** Both A and R are true and R is the correct explanation of A.	The green pigment traps light energy, which explains why chlorophyll is needed.
14	**(a)** Both A and R are true and R is the correct explanation of A.	Because light cannot bend around the opaque object, a dark region (shadow) forms behind it.
15	**(a)** Both A and R are true and R is the correct explanation of A.	Rotation turns each part of the Earth to and away from the Sun every 24 hours.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	The plant uses CO2 produced and gives oxygen (1); The candle can burn longer (1).	1+1
17	Angle of incidence is 0° (1); The ray retraces its path (1).	1+1
18	Moon reflects only a small fraction (1); Light is weak (1).	1+1
19	Lack of oxygen (1); They cannot respire (1).	1+1
20	The Sun is low in the evening (1); Rays fall at a smaller angle so the shadow stretches (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Hydrilla in water under funnel (1); Test tube collects gas (1); Control in dark (1).	1+1+1
22	i = r = 55° (1); Angle between rays 110° (1); Angle with mirror 35° (1).	1+1+1
23	15° in 1 hour (1); 1° in 4 minutes (1); 15° takes 1 hour (1).	1+1+1
24	Light is available at the top (1); Competition for sunlight (1); More photosynthesis (1).	1+1+1
	OR: Fix a stick vertically (1); Mark shadow lengths (1); Shortest shadow is at noon (1).	1+1+1
25	Earth's shadow is bigger (1); Moon's shadow small (1); Speeds (1).	1+1+1
	OR: Faster photosynthesis (1); More oxygen (1); Gas escapes (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) To use up (remove) the stored starch from the leaf (1). (ii) Iodine solution (1). (iii) Blue-black (1). (iv) The covered part does not turn blue-black (1), which shows that light is necessary for photosynthesis (starch formation) (1).	1+1+1+2

27	Fix a stick on the globe (1); Place lamp at the Sun's position (1); Tilt the globe (1); Measure shadows at two latitudes (1); Shadow longer at higher latitudes (1).	1+1+1+1+1
	OR: Set up three conditions (1); Same plant type (1); Starch test after some days (1); Only suitable one forms starch (1); Conclusions (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** nitrogen left by pulses	Rhizobium.
2	**(a)** red	Same colour.
3	**(d)** compare dates of nearest distance with seasons	January is nearest.
4	**(b)** NaOH absorbs carbon dioxide	No CO2.
5	**(c)** 1.6 m tall	Half.
6	**(b)** 182 days	Half.
7	**(c)** more glucose than used in respiration	Growth.
8	**(d)** double	Distance ratio 2:1.
9	**(b)** much nearer to the Earth	It appears equally large in the sky.
10	**(a)** mushrooms have no chlorophyll and absorb food from dead matter	Fungi are saprotrophs.
11	**(c)** black	Absence of light.
12	**(c)** incorrect	Phases are due to viewing angles.
13	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but guard cells do not explain why gases exchange occurs through stomata.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; the Moon only reflects sunlight, but the luminous nature of the Sun does not explain it.
15	**(d)** A is false but R is true.	Seasons are due to the tilt of the axis and revolution, not distance, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Earth rotates from west to east (1); Relative motion makes them seem to move opposite (1).	1+1
17	Spines and thick cuticle (1); Fewer stomata (1).	1+1
18	It reflects sunlight (1); Surface scatters light (1).	1+1
19	Sun stays above the horizon longer (1); Tilt effect (1).	1+1
20	No light for photosynthesis (1); Stored starch was used (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Water gives regular reflection (1); Light from headlights bounces back (1); An image of the lights is seen (1).	1+1+1
22	Study the Sun's corona (1); Test ideas about light (1); Predict future events (1).	1+1+1
23	Starch increases through the day (1); Photosynthesis exceeds respiration (1); Falls at night (1).	1+1+1
24	Initially 8 m (1); Now 3 m, so 6 m (1); Decrease 2 m (1).	1+1+1
	OR: First quarter about 8 March (1); Full moon about 16 March (1); Last quarter about 23 March (1).	1+1+1
25	They absorb carbon dioxide (1); They release oxygen (1); They trap dust on their leaves (1).	1+1+1
	OR: Reflect headlights (1); Improve visibility (1); Safety (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) A source of light, an opaque object and a screen (1). (ii) It becomes larger (1). (iii) Translucent (1). (iv) No (1), because clear glass is transparent and allows almost all the light to pass through (1).	1+1+1+2

27	Parasite or saprotroph (1); Check connection to living tissue (1); Observe on dead wood (1); Test for starch (1); Conclude (1).	1+1+1+1+1
	OR: Place object at several distances (1); Find image position by parallax with pin (1); Record distances (1); Compare in a table (1); Equal (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** it gains mass and grows	Balanced gas exchange.
2	**(c)** 50 cm	20 x 100/40.
3	**(a)** about 2.5 times	The Earth's shadow is large.
4	**(c)** saprotroph	Feeds on bread.
5	**(b)** light from a lamp reflecting from the book	Absorbs red.
6	**(c)** incorrect	Far side gets sunlight.
7	**(c)** high humidity	Humid air reduces.
8	**(a)** 2 m behind the mirror	Equal distance.
9	**(b)** every month	Alignment each month.
10	**(a)** photosynthesis is less than respiration	Low light.
11	**(a)** 50 m	40/0.8.
12	**(d)** the Sun is overhead	Equinox.
13	**(c)** A is true but R is false.	A is true; transpiration occurs mainly through stomata in leaves, so R is false.
14	**(c)** A is true but R is false.	A is true; a virtual image cannot be obtained on a screen, so R is false.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true but unrelated: being a satellite is not the reason it has no light of its own.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	5 cm (1); Virtual, erect and laterally inverted (1).	1+1
17	Slight offset from the axis (1); It lies close but not exactly (1).	1+1
18	Dissolves chlorophyll (1); Makes colour change clearer (1).	1+1
19	Image of the Sun on a screen (1); Do not look at the Sun directly (1).	1+1
20	Sun's position changes (1); Earth rotates (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	The Earth rotates from west to east (1); Places in the east face the Sun earlier (1); So sunrise time differs (1).	1+1+1
22	Large in shade (1); Small or spines in desert (1); Floating in ponds (1).	1+1+1
23	No light (1); Nothing to reflect (1); Need a source (1).	1+1+1
24	About 12 hours each (1); Little change (1); Sun nearly overhead (1).	1+1+1
	OR: Raise photosynthesis (1); Faster growth (1); Higher yields (1).	1+1+1
25	Height ratio = 15/30 (1); 0.5 x 6 (1); 3 cm (1).	1+1+1
	OR: Summer shadow shorter (1); Sun higher (1); Winter Sun lower (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) The tilt of the Earth's axis and its revolution around the Sun (1). (ii) Summer (1). (iii) About 21 March and 23 September (equinoxes) (1). (iv) Sydney is in the southern hemisphere (1), which is tilted away from the Sun in June, so it gets less direct sunlight and a shorter day (1).	1+1+1+2
27	First mirror shows front (1); Second mirror behind (1); Reflect image of back (1); Repeated images (1); Customer sees the back (1).	1+1+1+1+1

	OR: North of east in summer (1); South of east in winter (1); Due east on equinox (1); Tilt (1); Observation by markers (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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